

Activity 1

1. Stack and Queue :

Stack operations :

Push (54) $\rightarrow$ [54]

Push (52) $\rightarrow$ [54, 52]

Pop () $\rightarrow$ 52

Push (55) $\rightarrow$ [54, 55]

Push (62) $\rightarrow$ [54, 55, 62]

S = Pop () $\rightarrow$ 62

Queue operations :

enqueue (21) $\rightarrow$ [21]

enqueue (24) $\rightarrow$ [21, 24]

dequeue () $\rightarrow$ 21

enqueue (28) $\rightarrow$ [24, 28]

enqueue (32) $\rightarrow$ [24, 28, 32]

q = dequeue () $\rightarrow$ 24

S + q = 62 + 24

= 86

2. Postfix Expression :

105 + 606 / * 8 -

Calculation:

$$10 + 5 = 15$$

$$60 \div 6 = 10$$

$$15 \times 10 = 150$$

$$150 - 8 = 142$$

3. Postfix Expression

$$8 \ 2 \ 3 \wedge / 2 \ 3 * + 5 1 * -$$

After the first +

$$2^3 = 8$$

$$8 \div 8 = 1$$

$$2 \times 3 = 6$$

two element 1 & 6

4. Infix to Postfix

$$\text{Infix} : a + b * c - d \wedge e \wedge f$$

$$\text{Postfix} : abc + def \wedge \wedge - *$$

5. Stack POP sequence:

push(10)	10	-
push(20)	10, 20	-
POP	10	20
push(10)	10, 10	-
push(20)	10, 10, 20	-
POP	10, 10	20
POP	10	10
POP	empty	10
push(20)	20	-
POP	empty	20

POP Sequence : 20, 20, 10, 10, 20